

5.2.49

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Question

Construct a pair of tangents to a circle of radius 4 cm from a point P at a distance of 7 cm from its centre O .

Solution

Let the equation of the circle with centre $\mathbf{u}$ be

$$(\mathbf{x} - \mathbf{u})^\top (\mathbf{x} - \mathbf{u}) = r^2, \quad (1)$$

Given,

$$r = 4, \quad \|\mathbf{P} - \mathbf{u}\| = 7. \quad (2)$$

Let

$$\mathbf{u} = \mathbf{0}, \quad \mathbf{v} = \mathbf{P} - \mathbf{u}, \quad s = \mathbf{v}^\top \mathbf{v} = \|\mathbf{v}\|^2. \quad (3)$$

Define the orthonormal basis vectors as

$$\mathbf{v}_1 = \frac{\mathbf{v}}{\sqrt{s}}, \quad \mathbf{v}_2 = \frac{\mathbf{J}\mathbf{v}}{\sqrt{s}}, \quad \mathbf{J} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (4)$$

Let the unit normal to the tangent at the point of contact be expressed as

$$\frac{\mathbf{n}}{\|\mathbf{n}\|} = \alpha \mathbf{v}_1 + \beta \mathbf{v}_2. \quad (5)$$

Solution

The points of contact on the circle are given by

$$\mathbf{q}_i = \mathbf{u} + r \frac{\mathbf{n}_i}{\|\mathbf{n}_i\|}, \quad i = 1, 2. \quad (6)$$

Tangency condition:

For tangents from an external point $\mathbf{P}$ to the circle, we have

$$\mathbf{v}^\top \frac{\mathbf{n}}{\|\mathbf{n}\|} = r. \quad (7)$$

Substituting (5) into (7), and noting that $\mathbf{v} = \sqrt{s} \mathbf{v}_1$ and $\mathbf{v}^\top \mathbf{v}_2 = 0$, we get

$$\alpha \sqrt{s} = r \quad \Rightarrow \quad \boxed{\alpha = \frac{r}{\sqrt{s}}}. \quad (8)$$

Unit length condition:

$$\alpha^2 + \beta^2 = 1. \quad (9)$$

Solution

$$\beta^2 = 1 - \frac{r^2}{s} \quad \Rightarrow \quad \boxed{\beta = \pm \sqrt{1 - \frac{r^2}{s}}.} \quad (10)$$

Hence,

$$\boxed{\frac{\mathbf{n}}{\|\mathbf{n}\|} = \frac{r}{\sqrt{s}} \mathbf{v}_1 \pm \sqrt{1 - \frac{r^2}{s}} \mathbf{v}_2.} \quad (11)$$

$$\Rightarrow \boxed{\mathbf{q}_i = \mathbf{u} + \frac{r^2}{\sqrt{s}} \mathbf{v}_1 \pm r \sqrt{1 - \frac{r^2}{s}} \mathbf{v}_2.} \quad (12)$$

Solution

Given $r = 4$, $\sqrt{s} = 7$, $\mathbf{u} = \mathbf{0}$,

$$\frac{r^2}{\sqrt{s}} = \frac{16}{7} \quad (13)$$

$$r \sqrt{1 - \frac{r^2}{s}} = 4 \sqrt{1 - \frac{16}{49}} = \frac{4\sqrt{33}}{7}. \quad (14)$$

$$\mathbf{q}_1 = \begin{pmatrix} \frac{16}{7} \\ \frac{4\sqrt{33}}{7} \end{pmatrix}, \quad \mathbf{q}_2 = \begin{pmatrix} \frac{16}{7} \\ -\frac{4\sqrt{33}}{7} \end{pmatrix}. \quad (15)$$

C Code

```
#include <stdio.h>
#include <math.h>

// Tangent function to be called from Python
void tangent_points(double r, double px, double py, double *Ax,
    double *Ay, double *Bx, double *By) {
    double ox = 0.0, oy = 0.0;
    double d = sqrt((px - ox)*(px - ox) + (py - oy)*(py - oy));

    if (d <= r) {
        *Ax = *Ay = *Bx = *By = NAN;
        return;
    }

    double r2 = r*r;
    double d2 = d*d;
    double k = r2 / d2;
    double h = (r / d2) * sqrt(d2 - r2);
```

```
*Ax = k*px - h*py;  
*Ay = k*py + h*px;  
*Bx = k*px + h*py;  
*By = k*py - h*px;  
}
```



```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from numpy import linalg as LA
4
5 # Local imports
6 from libs.line.funcs import line_gen
7 from libs.conics.funcs import circ_gen
8
9 # --- Circle and external point ---
10 O = np.array([[0.0], [0.0]]) # Centre
11 r = 4.0 # Radius
12 P = np.array([[7.0], [0.0]]) # External point
13
14 # --- Tangent point calculation ---
15 u = P - O
16 d = float(LA.norm(u))
17 u_hat = u / d
```

Python Direct

```
# Rotation matrix for 90 degrees
R90 = np.array([[0, -1], [1, 0]])

# Tangent points formula
r2 = r**2
sqrt_term = np.sqrt(d**2 - r**2)

A = 0 + (r2/d)*u_hat + (r/d)*sqrt_term*(R90 @ u_hat)
B = 0 + (r2/d)*u_hat - (r/d)*sqrt_term*(R90 @ u_hat)

# --- Generate required curves ---
x_circ = circ_gen(0, r)
x_tan1 = line_gen(P, A)
x_tan2 = line_gen(P, B)
x_OP = line_gen(0, P)

# --- Plotting ---
plt.plot(x_circ[0, :], x_circ[1, :])
```

```
plt.plot(x_tan1[0, :], x_tan1[1, :], color='orange' )
plt.plot(x_tan2[0, :], x_tan2[1, :], color='green' )
plt.plot(x_OP[0, :], x_OP[1, :], 'k--')

# --- Mark and label points (rounded neatly) ---
points = {'O': O, 'P': P, 'A': A, 'B': B}
for name, pt in points.items():
    x, y = float(pt[0]), float(pt[1])
    plt.scatter(x, y, s=45, zorder=5)
    plt.text(x + 0.25, y + 0.25,
             f'{name}({round(x,2)}, {round(y,2)})',
             fontsize=9, ha='center')
```

```
# --- Axes setup ---
ax = plt.gca()
ax.spines['top'].set_color('none')
ax.spines['right'].set_color('none')
ax.spines['left'].set_position('zero')
ax.spines['bottom'].set_position('zero')

# --- Style ---
plt.axis('equal')
plt.grid(True)
plt.xlabel('$x$')
plt.ylabel('$y$')
plt.title('Tangents from an External Point to a Circle')
plt.legend(loc='upper right')
plt.show()
```

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt
from libs.line.funcs import line_gen
from libs.conics.funcs import circ_gen

# Load the shared library
lib = ctypes.CDLL('./libtangent.so')

# Define argument and return types
lib.tangent_points.argtypes = [
    ctypes.c_double, ctypes.c_double, ctypes.c_double,
    ctypes.POINTER(ctypes.c_double), ctypes.POINTER(ctypes.
        c_double),
    ctypes.POINTER(ctypes.c_double), ctypes.POINTER(ctypes.
        c_double)
]
```

```
# Inputs
r = 4.0
P = np.array([[7.0], [0.0]])

# Prepare output containers
Ax = ctypes.c_double()
Ay = ctypes.c_double()
Bx = ctypes.c_double()
By = ctypes.c_double()

# Call the C function
lib.tangent_points(r, float(P[0]), float(P[1]),
                  ctypes.byref(Ax), ctypes.byref(Ay),
                  ctypes.byref(Bx), ctypes.byref(By))

# Convert results to numpy arrays
O = np.array([[0.0], [0.0]])
A = np.array([[Ax.value], [Ay.value]])
```

```
B = np.array([[Bx.value], [By.value]])

print(Tangent Points from C library:)
print(fA({A[0,0]:.2f}, {A[1,0]:.2f}))
print(fB({B[0,0]:.2f}, {B[1,0]:.2f}))
# Plot using your libs
x_circ = circ_gen(0, r)
x_tan1 = line_gen(P, A)
x_tan2 = line_gen(P, B)
x_OP = line_gen(0, P)
plt.plot(x_circ[0, :], x_circ[1, :], label='Circle')
plt.plot(x_tan1[0, :], x_tan1[1, :], color='orange', label='
    Tangent 1')
plt.plot(x_tan2[0, :], x_tan2[1, :], color='green', label='
    Tangent 2')
plt.plot(x_OP[0, :], x_OP[1, :], 'k--', label='$OP$')
# Mark and label points
points = {'O': 0, 'P': P, 'A': A, 'B': B}
for name, pt in points.items():
```

```
x, y = float(pt[0]), float(pt[1])
plt.scatter(x, y, s=45, zorder=5)
plt.text(x + 0.25, y + 0.25, f'{name}({round(x,2)}, {round(y,2)})',
         fontsize=9, ha='center')
# -style axes
ax = plt.gca()
ax.spines['top'].set_color('none')
ax.spines['right'].set_color('none')
ax.spines['left'].set_position('zero')
ax.spines['bottom'].set_position('zero')
plt.axis('equal')
plt.grid(True)
plt.xlabel('$x$')
plt.ylabel('$y$')
plt.title('Tangents from External Point (using C library)')
plt.legend(loc='upper right')
plt.show()
```


Plot

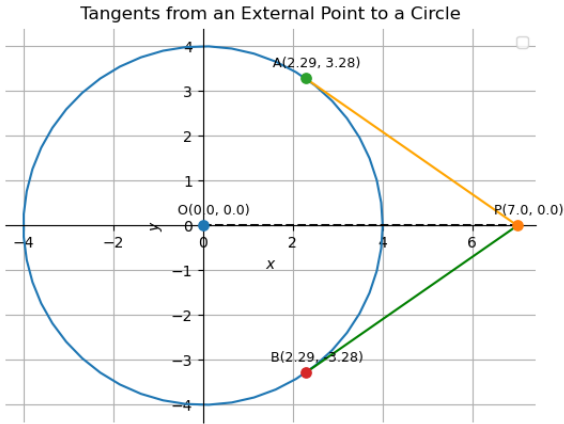


Figure: